MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 5J26

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, tunable frequency range 1, 220 to 1, 350 MHz, external magnet, rated peak power output = 400 kw

ABSOLUTE RATINGS:

Parameter:	Ef	if(surge)	tk	tpc	prr	Du	epy	ib	Pi	VSWR	T(anode)	Tuner torque
Unit:	V	a	sec	μs	pps	---	kv	a	W	---	°C	in.-oz
Maximum:	26.0	4	300	6.0	1,000	0.0025	34	55	1,725	1.5:1	125	128
Minimum:	--- (Note 7)	---	180	1.0	---	---	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any
Cathode: Unipotential, oxide coated
Weight: 20 pounds (approximately)(9.07 kg)

TEST CONDITIONS:

Parameter:	Ef	tpc	trv	tfc	Du	H	VSWR	Ib
Unit:	V	μs	μs	μs	---	Gauss	---	mAdc
Test condition 1:								
Maximum:	---	1.10	0.26	---	---	---	1.10:1	---
	15.5	---	---	0.25	0.001	1,400	---	46
Minimum:	--- (Notes 7 & 12)	0.90	0.24	---	---	---	---	---
Test condition 2:								
Maximum:	---	4.60	0.50	---	---	---	1.10:1	---
	15.5	---	---	1.20	0.001	1,400	---	52
Minimum:	--- (Notes 7 & 12)	3.80	0.30	---	---	--- (Note 10)	---	---

Frequency
F1 = 1,220 MHz
F2 = 1,285 MHz
F3 = 1,350 MHz

GENERAL:

Qualification - Required

Ⓖ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
Ⓖ 1047	Low-temperature operation	---	2	tk = 300 sec (max); F2	Po	320	---	W
Ⓖ 4027	Temperature coefficient	9	2	T(anode) = 70° to 100°C; F1, F2, F3	ΔF/ΔT	---	0.03	MHz/°C
Ⓖ 1042	Shock	---	---	Test condition G	---	---	---	---
	Quality conformance inspection, part 1	11	---					
4289	Heater current	7, 12	---	Ef = 23.5 V	If	2.00	2.40	A
4223	Mechanical tuning range	---	1	Upper limit Lower limit	F F	1,350 ---	--- 1,220	MHz MHz
4003	Pressurizing	---	---	30 psia	---	---	---	---
4306	Pulse voltage	---	1	F1, F2, F3	epy	26.5	31.5	kv
4250	Power output	---	1	F1, F2, F3	Po	400	---	W
4308	RF bandwidth	2,3 2,3	1 2	F1, F2, F3 F1, F2, F3	BW BW	--- ---	2.0/tpc 2.5/tpc	MHz MHz
4315	Starting stability	1,4	2	VSWR = 1.5:1; F1	MP	---	2.0	%
	Quality conformance inspection, part 2							
1031	High-frequency vibration	---	---	No voltages applied	---	---	---	---
4223	Tuner operating torque	5,6	---		Torque	---	128	in.-oz
4310	Frequency pulling factor	---	1	VSWR = 1.5:1; F1, F2, F3	ΔF	---	5.0	MHz
4311	Frequency pushing factor	8	1	F1, F2, F3	ΔF	---	0.5	MHz
Ⓖ 4315	Stability	4	1	VSWR = 1.5:1; F1	MP	---	1.0	%
	Quality conformance inspection, part 3							
4551	Life test	---	2	Group D	t	500	---	hours
---	Life-test end points:							
4250	Power output	---	1	F1, F2, F3	Po	320	---	W
4308	RF bandwidth	2,3	2	F1, F2, F3	BW	---	4.0/tpc	MHz

NOTES:

1. The starting stability test shall be performed on 100 percent of the submitted lot.
2. Frequency skipping or unstable operation may be encountered at some phase positions when the mismatch occurs at the end of a line 8 meters long, or longer.
3. The rf bandwidth shall be within the limits specified when a VSWR of 1.5:1 is introduced in the load at a distance of approximately 0.5 meter from the magnetron coupling flange.
4. The missing pulses (MP), due to any causes, are considered to be missing if the rf energy is less than 70 percent of the normal energy level in the frequency range of 1,220 to 1,350 MHz.
5. This test shall be conducted over a temperature range of -60° to +100°C inclusive. The tube shall not be operating during this test.

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NOTES: -Continued

6. The tuner drive mechanism shall not be set against either mechanical stop.

7. The heater surge current shall not exceed 4.0 amperes.

8. The anode current shall be varied between 46 ±10 percent amperes peak at a minimum rate of 50 cycles per second. The ΔF measurement is the difference between the frequency extremes measured.

9. The temperature is to be measured at the point indicated on figure 1 (side view).

10. The north seeking pole shall be adjacent to the cathode terminal.

11. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

12. A tk of 180 seconds (min) with an EF of 23.5 volts is required before test.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

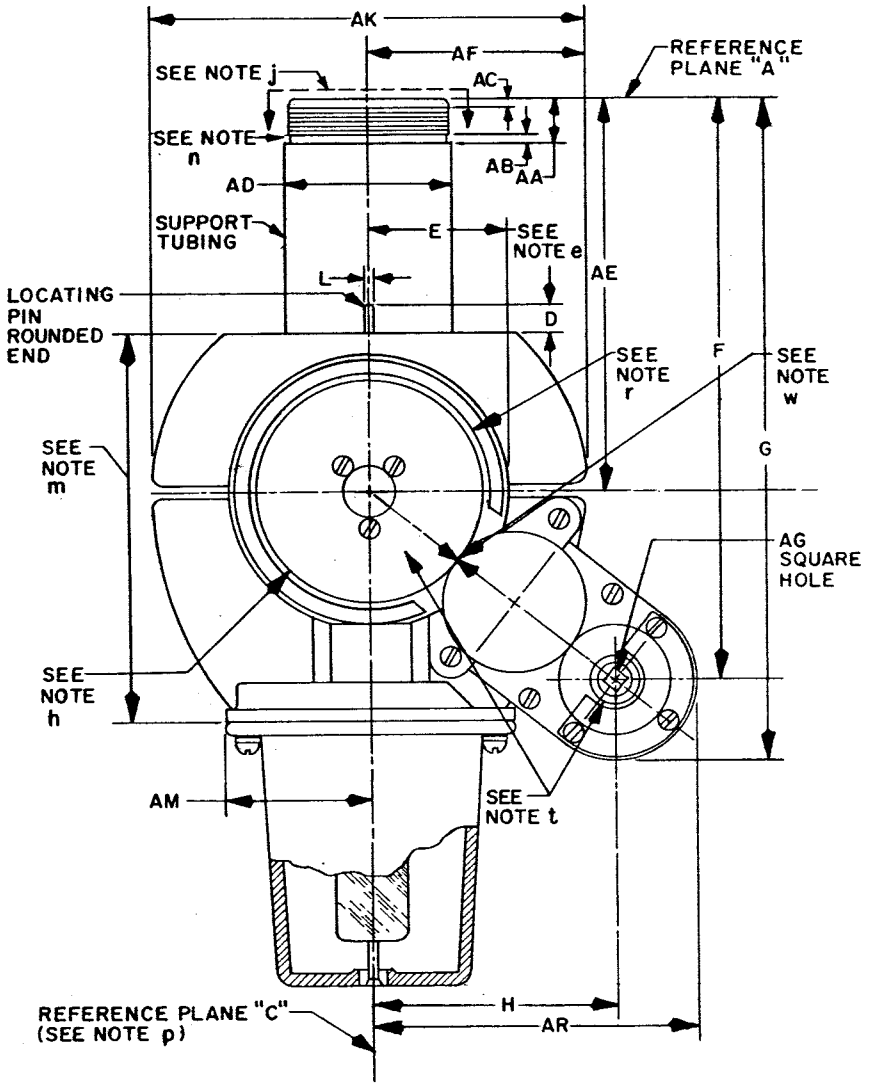
User activities:
Army - AV, ME, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2968)

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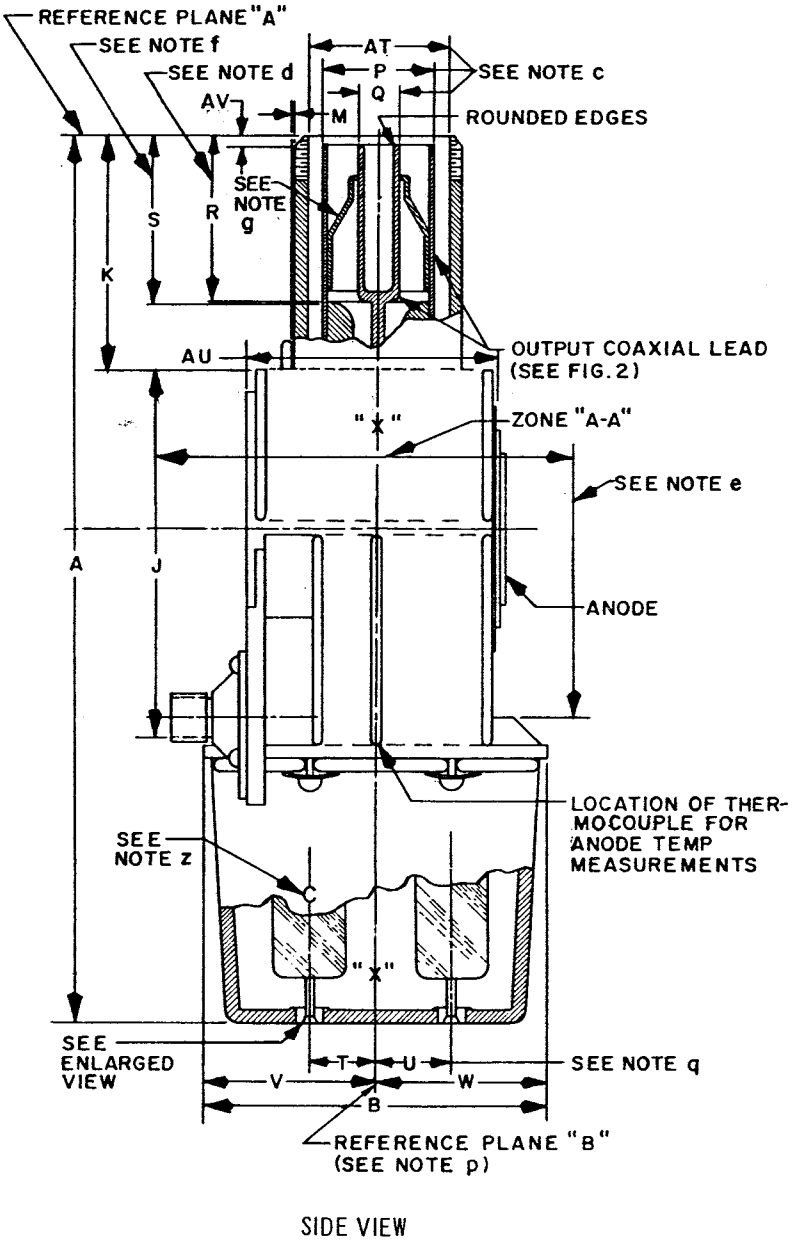


FRONT VIEW

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
D	.359 (9.12)	.391 (9.93)
L	.122 (3.10)	.128 (3.25)
AA	.531 (13.49)	.625 (15.88)
AD	2.307 (58.60)	2.312 (58.72)
AE	5.406 (137.31)	5.531 (140.49)
Quality conformance inspection, part 2		
F	7.812 (198.42)	8.188 (207.98)
G		9.188 (233.38)
H		3.375 (85.73)
AB	.125 (3.18)	
AC	.062 (1.57)	
AF		3.000 (76.20)
AG	.248 (6.30) x .248 (6.30)	.252 (6.40) x .252 (6.40)
AK		6.000 (152.40)
AM		2.000 (50.80)
AR		4.500 (114.30)
Quality conformance inspection, part 3		
E	2.375 (60.33)	

FIGURE 1. Outline drawing of electron tube type 5J26.

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SIDE VIEW

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
K	3.047 (77.39)	3.062 (77.77)
P	1.567 (39.80)	1.587 (40.31)
Q	.554 (14.07)	.574 (14.58)
AT	1.927 (48.95)	1.935 (49.15)
AU	3.450 (87.63)	3.560 (90.42)
AV	.080 (2.03)	.115 (2.92)
Quality conformance inspection, part 2		
A		12.500 (317.50)
B		4.750 (120.65)
J	4.875 (123.83)	
M		.010 (.25)
R	2.328 (59.13)	
S	2.344 (59.54)	
V		2.297 (58.34)
W		2.453 (62.31)
X	.012 (.30)	
AP	.278 (7.06)	.284 (7.21)
Quality conformance inspection, part 3		
T	.904 (22.96)	
U	1.036 (26.31)	
AN	.025 (.64)R	
AQ	.593 (15.06)	

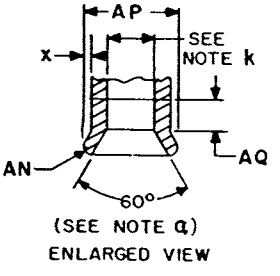
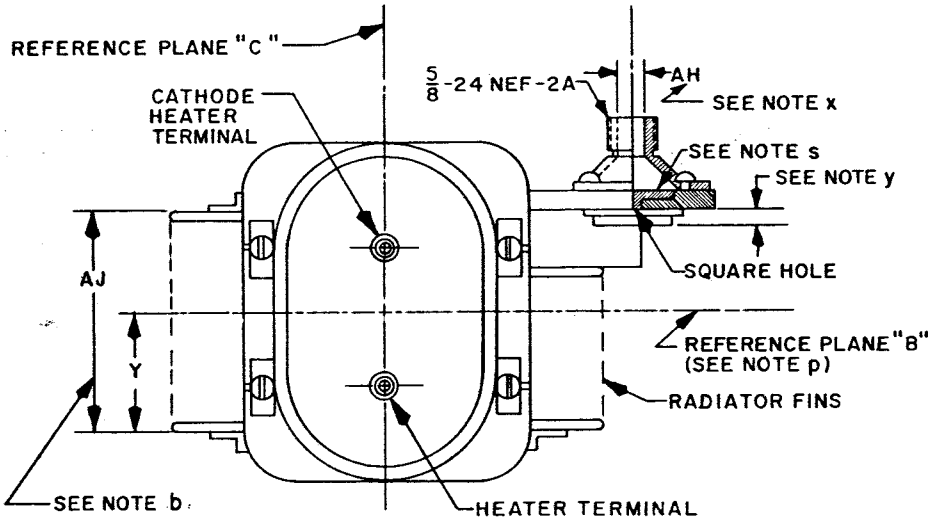


FIGURE 1. Outline drawing of electron tube type 5J26 - Continued.

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BOTTOM VIEW

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
Y		1.562 (39.67)
AH	.373 (9.47)	.377 (9.58)
AJ	2.625 (66.58)	3.000 (76.20)

FIGURE 1. Outline drawing of electron tube type 5J26 - Continued.

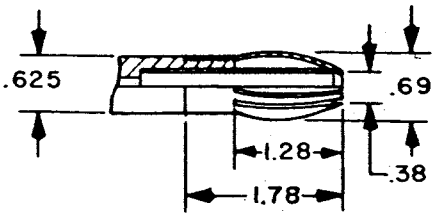
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NOTES:

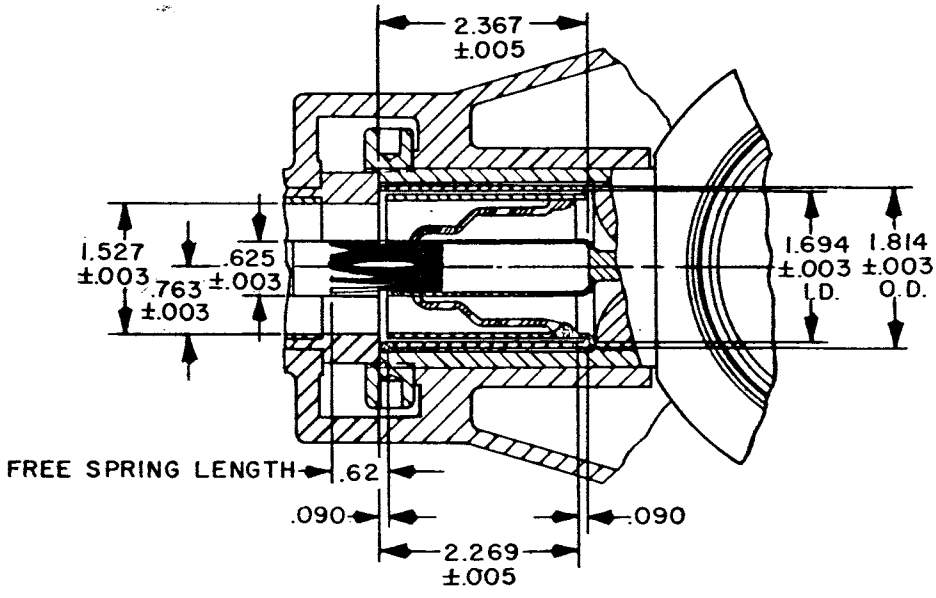
a. This countersink shall be capable of accepting a cone having a 60 degree included angle and base diameter of .234 (5.94 mm) and shall reject a similar cone having a base diameter of .254 (6.45 mm).
b. This dimension applies to the tip of the radiator fins.
c. The concentricity of these dimensions with respect to each other shall be considered satisfied if they meet the following conditions: Concentric tubes and plug measuring 2.315 (58.80 mm)-inside diameter, 1.926 (48.92 mm)-outside diameter, 1.600 (40.64 mm)-inside diameter and .530 (13.46 mm) diameter shall pass freely into or over these parts to the distances indicated in notes d and f.
d. Depth to which the 1.926 (48.92 mm)-outside diameter, 1.600 (40.64 mm)-inside diameter tube shall penetrate (see note c).
e. Zone 'A - A' shall be capable of being inserted between two parallel planes 3.640 (92.41 mm) apart which are equidistant from the axis of a 2.315 (58.80 mm) diameter tube into which the support tube is inserted. The locating pin shall fall between parallel planes .150 (3.81 mm) apart equidistant from the 'X - X' axis and normal to the above noted planes 3.640 (58.80 mm) apart.
f. Depth to which the .530 (13.46 mm) diameter plug shall penetrate (see note c).
Ⓖg. Contour of glass or ceramic may vary from that shown within limiting bounds defined by concentric cylinders and plugs referred to in note c.
h. Gears shall be underflush with respect to gear housing.
j. The opening in the support tubing shall be protected by a dust cover when not in use.
k. Jack holes .169 ±.005 x .594 (4.29 ±.13 x 15.08 mm) deep minimum not including tapered section.
m. All surfaces in this area shall have a heat resistant finish except gears, gear bearing surfaces, gear supports and gear housings.
n. 2.312-16N.S. 5-full threads minimum.
Maximum major diameter 2.3120 (58.72 mm).
Minimum major diameter 2.2980 (58.37 mm).
Maximum pitch diameter 2.2714 (57.69 mm).
Minimum pitch diameter 2.2629 (57.47 mm).
Maximum minor diameter 2.2353 (56.77 mm).
p. Reference plane B shall be parallel to either face of the anode. Reference planes B and C shall be at right angles to each other and intersect at the axis of the support tubing.
q. The jack shall be concentric with respect to the hole in the boot within .062 (1.57 mm). The centers of the holes in the boot shall be within a radius of .062 (1.57 mm) of the specified centers.
r. These two gears rotate clockwise when increasing frequency.
s. Complete frequency range covered in approximately 9-1/2 turns of driving gear.
t. Visible movable parts shall be free from corrosion, paint or other surface imperfections. The tuning mechanism shall operate over the entire mechanical range with a maximum torque of 8 inch-pounds applied to the driving gear.
u. Lubricant shall appear on visible bearing surfaces of all movable parts.
v. All screws shall be nonmagnetic, noncorrosive material, or shall have noncorrosive heat resistant finish. Screw heads may be hexagon head, hexagon socket, or straight slotted.
w. Matched arrows indicate approximate midband frequencies.
x. This round hole shall be concentric with the square hole within .003 (.08 mm).
y. In this zone there shall be a concentric clearance of .360 (9.14 mm) diameter centered on axis of rotation of driver gear.
Ⓖz. Cathode-heater lead shall be marked with a letter "C" at either the top or side of the protective boot.

FIGURE 1. Outline drawing of electron tube type 5J26 - Continued.

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THE CONNECTOR SHOULD BE CONSTRUCTED TO REQUIRE A FORCE OF BETWEEN 6 AND 12 POUNDS TO ENGAGE WITH THE TUBE. CONNECTORS CONSTRUCTED OF .015 THICK HALF HARD BERYLLIUM COPPER STRIP (A.S.T.M. B-120 1/2 H), HAVING 12 SEGMENTS SEPARATED BY 1/32 SAWCUTS, HAVE BEEN FOUND TO MEET THIS REQUIREMENT.



INCHES	MM
.003	.08
.005	.13
.015	.38
.090	2.29
.38	9.65
.62	15.75
.625	15.88
.69	17.53
.763	19.38
1.28	32.51
1.527	38.79
1.694	43.03
1.78	45.21
1.814	46.08
2.269	57.63
2.367	60.12

NOTE:
THE DIMENSIONED CYLINDRICAL SURFACES SHALL BE CONCENTRIC WITHIN .003

FIGURE 2. Output coupling.

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